The encrypted bytes are decrypted using the same nonce and key. Data encrypted in this way is normally Base64 encoded (for binary safety) before sending over a network, and the nonce is transmitted as part of the message (it exists simply to make the message unique and prevents malicious re-transmission by an attacker; it serves no additional purpose in obfuscating the message). Once the recipient has (if necessary) decoded and extracted the nonce, decryption is very straightforward:

```
$decrypted_message = sodium_crypto_secretbox_open(
    $encrypted_message,
    $nonce,
    $secret
);
print $decrypted_message;
```

```
Output:
WE RIDE AT DAWN.
```

Public Key

For a key-pair-based implementation, we use sodium's `sodium_crypto_box()` as our encryption function and create encryption key pairs by combining the private key of one party with the public key of the other. First, we'll create public/private keys (sodium generates ED25519-based keys, in case you're wondering):

```
$alicePair = sodium_crypto_box_keypair();
$pubAlice = sodium_crypto_box_publickey($alicePair);
$privAlice = sodium_crypto_box_secretkey($alicePair);

$bobPair = sodium_crypto_box_keypair();
$pubBob = sodium_crypto_box_publickey($bobPair);
$privBob = sodium_crypto_box_secretkey($bobPair);
```

`sodium_crypto_box_keypair()` creates both keys, unified into a single byte sequence. We then extract the public and private sections using `sodium_crypto_box_publickey()`, etc. Bob gets his own pair of keys using the same method. Next, we generate the encryption keys: